Christopher R. Slade

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Education

Ph.D. Computer Science, University of Hawaii, 2008-2012, 2023-2026
Dissertation: Enhancing Mobile Health Sensing with Authorization Techniques, Persistent Reminders, and Adaptive EMA Timing
Graduation: Aug. 2026

M.S. Computer Science, Brigham Young University, 2005
Thesis: On-Disk Sequence Cache (ODSC): Using Excess Disk Capacity to Increase Performance.

B.S. Computer Science, Brigham Young University, 2002, *Summa Cum Laude*

Research Experience

Hawaii Digital Health Lab — 2022–Present
Peter Washington — University of Hawaii - Manoa
Mobile Health Sensing

Laboratory for Interactive Learning Technologies — 2008–2012
Daniel D Suthers — University of Hawaii - Manoa
Online Context Mapping Technology, Threaded Discussions

Performance Evaluation Lab — 2003–2007
Kelly Flanagan — Brigham Young University
Disk Drive Performance, I/O Systems, Memory Management

Teaching Experience

Associate Professor, Aug. 2007–Present
Brigham Young University-Hawaii
Courses: Algorithms and Complexity, Artificial Intelligence, Computational Theory, Computer Organization, Beginning and Object-Oriented Programming, Discrete Mathematics, Data Structures, Operating Systems, Robotics, Programming Languages, Web Application Development, Advanced Web Application Development, Mobile Application Development. Curriculum design and development, authoring several curriculum change proposals.

Graduate Student Instructor (unpaid), Jan. 2007–Apr. 2007
Brigham Young University
Course: Programming Languages

Teaching Assistant, Jan. 2000–Dec. 2003
Brigham Young University
Courses: Computer Organization, Advanced Computer Architecture

Employment

Brigham Young University-Hawaii — Laie, HI
Associate Professor — August 2007–Present
Teaching, Research, and Curriculum Design

Independent Contractor — Laie, HI
Software Developer and Consultant — July 2015–Present
Provided software consulting and development services, using the following tools and technologies: Ruby on Rails, Laravel, Bootstrap, TailwindCSS, Swift, Node, Express, React.js, Vue.js, Angular.js, Docker, Amazon Web Services (AWS), MongoDB.

College of Health & Human Performance — BYU, Provo, UT
Web Programmer — May 2002–Aug. 2003
Designed and developed several web-based applications including a library catalog and an intramural scheduling & registration application used by more than 10,000 participants at a time.

Novell — Provo, UT
Competitive Analyst — May 2001–Aug. 2001
Set up a competitive research lab and compared four different collaboration applications. Developed a Java servlet to connect a partner's product to a web portal.

Agilent Technologies — Colorado Springs, CO
Web Programmer — May 2000–Aug. 2000
Designed and developed a web-based collaboration application, to help the marketing and development departments share product information and marketing data.

Publications

Kargarandehkordi A, Jaiswal A, Banerjee A, Qian Y, Slade CR, Sun Y, Islam T, Tadesse HB, Kostrinsky-Thomas A, Park C, Sarkar U, Khoong EC, Nguyen N, Xu X, Phillips KT, Benzo RM, Aguilera A, Zhang H, Doshi-Velez F, Washington P. Personalized modeling of stress and blood pressure reactivity using mobile health data. npj Mental Health Research. 2026 May. doi: 10.1038/s44184-026-00210-9.

Slade C, Sun Y, Chao WC, Chen CC, Benzo RM, Washington P. Current Challenges and Opportunities for Active and Passive Data Collection for Mobile Health Sensing: A Scoping Review. JAMIA Open. 2025 Jul. doi: 10.1093/jamiaopen/ooaf025.

Sun Y, Jaiswal A, Slade C, Phillips KT, Benzo RM, Washington P. Associations Between Social Determinants of Health and Adherence in Mobile-Based Ecological Momentary Assessment: Scoping Review. J Med Internet Res. 2025 Sep. doi: 10.2196/69831.

Slade C, Benzo RM, Washington P. Design Guidelines for Improving Mobile Sensing Data Collection: Prospective Mixed Methods Study. J Med Internet Res. 2024 Nov. doi: 10.2196/55694.

Li S, Fan C, Kargarandehkordi A, Sun Y, Slade C, Jaiswal A, Benzo RM, Phillips KT, Washington P. Monitoring Substance Use with Fitbit Biosignals: A Case Study on Training Deep Learning Models Using Ecological Momentary Assessments and Passive Sensing. AI. 2024 Nov. doi: 10.3390/ai5040131.

Kargarandehkordi A, Slade C, Washington P. Personalized AI-Driven Real-Time Models to Predict Stress-Induced Blood Pressure Spikes Using Wearable Devices: Proposal for a Prospective Cohort Study. JMIR Res Protoc. 2024 Mar 25;13:e55615. doi: 10.2196/55615. PMID: 38526539; PMCID: PMC11002732.

Sun Y, Kargarandehkordi A, Slade C, Jaiswal A, Busch G, Guerrero A, Phillips KT, Washington P. Personalized Deep Learning for Substance Use in Hawaii: Protocol for a Passive Sensing and Ecological Momentary Assessment Study. JMIR Res Protoc. 2024 Feb 7;13:e46493. doi: 10.2196/46493. PMID: 38324375; PMCID: PMC10882478.

Wolthuis, Stuart L and Slade, Christopher, "Liberating Legacy System Data with Rails, Intelligent Use of Conflict Data with Automated Class Scheduling Tools.", Information Systems Education Journal, Volume 18, Pg. 12-21, 2020

Conference Papers

Kargarandehkordi A, Banerjee A, Jaiswal A, Qian Y, Slade CR, Sun Y, Flynn A, Hai M, Singh R, Nguyen N, Xu X, Phillips KT, Benzo RM, Aguilera A, Doshi-Velez F, Washington P. Person-Specific Blood Pressure Spike Modeling Using Wearable Data Across Multiple Temporal Aggregation Scales. In: Schmorrow DD, Fidopiastis CM, editors. Augmenting Cognition in the AI-Accelerated Era. Cham: Springer Nature Switzerland; 2026. p. 223–237. doi: 10.1007/978-3-032-29548-4_15.

Slade, Christopher R. Suthers, D. D.(2011). Using Interactive Graphical Tools to overcome the "Collage Effect" in Collaborative Learning Environments. In Proceedings of the 44th Hawai'i International Conference on the System Sciences (HICSS-34), January 5-7, 2011, Po'ipu, Hawai'i (CD-ROM). New Brunswick: Institute of Electrical and Electronics Engineers, Inc. (IEEE).

Thesis/Dissertation

Enhancing Mobile Health Sensing with Authorization Techniques, Persistent Reminders, and Adaptive EMA Timing. Doctoral Dissertation. University of Hawaii at Manoa. Aug. 2026.

On-Disk Sequence Cache (ODSC): Using Excess Disk Capacity to Increase Performance. Master's Thesis. Brigham Young University. Dec.2005. http://contentdm.lib.byu.edu/ETD/image/etd1020.pdf

Presentations

Kevin Kimball, Christopher Slade "Accounting Balls: An iPhone Game to Teach Debits and Credits" 7th Annual Conference on Teaching and Learning in Accounting, Atlanta, Georgia (August 2014)

Speeches

Christopher Slade "Wilt Thou Not Place a Stumbling Block in My Way" University-wide Devotional, Brigham Young University - Hawaii (March 31, 2020) https://speeches.byuh.edu/devotionals/wilt-thou-not-place-a-stumbling-block-in-my-way

Service

Assessment Chair, Department of Computer and Information Systems, Brigham Young University-Hawaii, January 2026-Present.

Promotion Review Committee, Brigham Young University-Hawaii, August 2024-Present.

Tutoring Supervisor, Department of Computer and Information Systems, Brigham Young University-Hawaii, January 2008-August 2009, August 2016-August 2024.

Academic Exceptions Committee, Brigham Young University-Hawaii, August 2015-August 2019.

Faculty Advisory Committee (Faculty Senate), Brigham Young University-Hawaii, August 2013-May 2016.

Computer and Information Sciences Professional Society Faculty Advisor, Department of Computer and Information Systems, Brigham Young University-Hawaii, August 2011-2012.

Computer Science Professor Search Committee Chair, Department of Computer and Information Systems, Brigham Young University-Hawaii, September 2008.

Computer Science Program Lead, Department of Computer and Information Systems, Brigham Young University-Hawaii, August 2008-Dec. 2009.

Scholarship Committee Member, Department of Computer and Information Systems, March 2008.

Industrial Relations Committee Member, Department of Computer and Information Systems, Brigham Young University-Hawaii, January 2008-April 2008.

Professional Societies and Organizations

Member of the Association for Computing Machinery (ACM)

Miscellaneous

Served a two-year mission for the Church of Jesus Christ of Latter-day Saints in Brazil.

Fluent in Portuguese.

Revised: June 2026